

Where the wine isn't

A world map of land fit for vineyards, and what a shrinking market would pay for its wine

Land fit for vineyards lies idle across Turkey, Ukraine, southern Russia and Iran. We map it at 9km resolution, test the map on continents the model never saw, and price what the land forgoes in a world that drinks less wine every year

Climate 1991-2020; vineyards as mapped in September 2026; wine-grape area to 2023; market data to 2025

115m

Hectares of farmland, without vines, that pass our strict test of vineyard suitability. The world grows wine grapes on 4.5m

0.96

Accuracy (AUC) at telling vineyard land from other land in countries the model never saw; a textbook temperature rule scores 0.92

32% vs 3%

21 countries that plant little of their suitable land: their share of it, against their share of the world's wine grapes

\$17bn

Wine sales a year those countries forgo against their share of suitable land, at export prices (range \$9bn-20bn)

Where wine is made owes as much to history as to geography. Monks, merchants, colonists and migrants decided where vines went, and laws and religions decided where they could not. This report asks a narrower question: if wine were grown wherever the land suits it, where would it be grown, and what are the places that grow almost none of it giving up?

We answer in three steps. First, we build a model of what vineyard land looks like, from the climate, water and soil under the vineyards of 26 wine countries, and test it on continents it was not shown. Second, we score every 9km square of the world's land and measure the farmland that looks like vineyard land but has no vines. Third, we value what the countries that plant least of it forgo, in a world market that has been shrinking since 2018. A fourth section asks whether the same land data can predict how good a region's wine is.

The data

Every input was downloaded, none typed by hand, and all are open. Climate is TerraClimate's monthly normals for 1991-2020, which give temperatures, rain, sunshine, evaporation and the water deficit at about 4km; we average them to a 5 arc-minute grid, about 9km at the equator, which yields 2.1m land cells between 60°S and 72°N. Soil comes from SoilGrids 2.0 (pH, clay, sand, organic carbon, nitrogen, stones and nutrient capacity at 15-30cm). Irrigation comes from the FAO's Global Map of Irrigation Areas. Cropland is the area of all 173 crops in CROPGRIDS for 2020.

The hardest input is the map of today's vineyards. CROPGRIDS includes grapes, but for most countries it inherits a census-based map from 2000 that spreads each district's grapes across the district. It shows no grapes at Stellenbosch or in Marlborough and puts South Africa's vineyards in the Karoo. A first version of our model trained on it failed its own tests in Africa and Oceania. We replaced it with OpenStreetMap, the open map of the world, whose volunteers have drawn 872,520 vineyards covering 3.46m hectares, about half the world's vine area. For South Africa it gives 119,887 hectares, against 119,471 from an independent download of the same map. We use OpenStreetMap for a country when it maps at least half of the country's grape area, and CROPGRIDS only where its sources are fine-grained. Eleven wine countries, including Spain and Italy, meet neither test; they are scored by the model but not used to train it.

Wine-grape areas by country for 2000, 2010, 2016 and 2023 come from the University of Adelaide's database by Kym Anderson, Signe Nelgen and Germán Puga; production, prices and consumption from Kym Anderson and

Vicente Pinilla's database of global wine markets; and the latest world figures from the OIV's report on 2025. Ratings come from X-Wines, 21m consumer ratings of 100,646 wines from 62 countries.

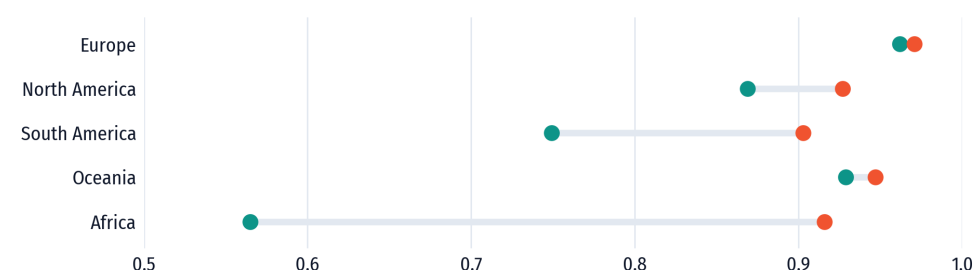
Reading the land

The model is a set of gradient-boosted decision trees. It learns to tell the 6,791 vineyard cells in the 26 training countries from their other land, using 25 measures of climate, water and soil. Two choices matter. Countries that grow grapes mainly for the table, for raisins or hardly at all are never used as evidence that a place is unsuitable; the absence of vines in Iran or Turkey says more about law, faith and markets than about land. And every training country is scored by models that never saw it, so its scores are tests, not fits.

Finding vineyards it was never shown

Accuracy at telling vineyard land from other land on a continent hidden from the model (AUC: 0.5 is a coin toss, 1 is perfect)

Our model Textbook temperature rule



Asia is left out: only 69 vineyard cells (Armenia) had a trustworthy map
Source: The Signum Average

The main test hides a whole continent. Shown no African data, the model ranks a South African vineyard cell above an ordinary South African cell 92% of the time (an area under the curve, or AUC, of 0.92). A textbook rule, which scores land by how close its growing-season temperature is to 17°C, the middle of the 12-22°C band associated with quality wine, manages 0.57. The model beats the rule on every continent that can be tested, by a wide margin in Africa and South America and narrowly in Europe, where temperature alone explains much. In each hidden continent the best-scoring tenth of land holds 82-99% of the vine hectares. Asia cannot be tested: only Armenia's 69 vineyard cells had a trustworthy map.

Known places Score of the 9km cell at each place; wine-country cells scored by models that never saw that country

	Score	Land in wine countries scoring lower	Test passed
Stellenbosch, South Africa	0.98	100%	strict
Marlborough, New Zealand	0.93	98%	strict
Central Otago, New Zealand	0.28	89%	neither
Napa, United States	0.94	98%	strict
Willamette, United States	0.23	89%	neither
Mendoza, Argentina	0.65	93%	generous
Maipo, Chile	0.99	100%	strict
Barossa, Australia	0.99	100%	strict
Bordeaux, France	0.98	100%	strict
Champagne, France	0.98	100%	strict
Rioja, Spain	0.98	100%	strict
Mosel, Germany	0.91	98%	strict
Sussex, England	0.27	89%	neither
Kakheti, Georgia	0.97	99%	strict
Ningxia, China	0.57	92%	generous
Cappadocia, Turkey	0.26	89%	neither
Shiraz, Iran	0.16	88%	neither
Urmia, Iran	0.92	98%	strict
Qazvin, Iran	0.93	98%	strict
Samarkand, Uzbekistan	0.93	98%	strict
Bekaa, Lebanon	0.63	93%	generous
Meknes, Morocco	0.41	91%	neither
Nashik, India	0.00	66%	neither
Sahara (control)	0.00	67%	neither
Amazon (control)	0.00	67%	neither
Siberia (control)	0.00	33%	neither
Kansas (control)	0.33	90%	neither
Guangdong (control)	0.02	78%	neither
Iowa (control)	0.01	71%	neither

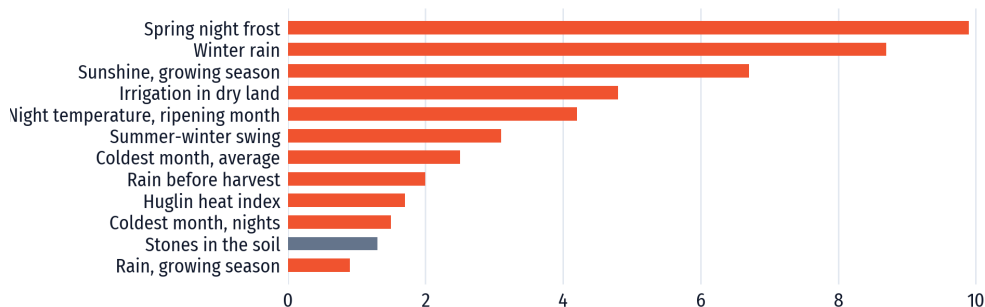
Strict test: score of 0.79 or more; generous: 0.49 or more. Source: The Signum Average

The second test looks up well-known places. Stellenbosch, Marlborough, Napa, Maipo, the Barossa, Bordeaux, Champagne, Rioja, the Mosel and Kakheti all score in the top 2-3% of land in wine countries, and so do Iran's grape districts of Urmia and Qazvin and Uzbekistan's Samarkand. Mendoza, Ningxia and the Bekaa pass only the generous test. Seven places fail both: Central Otago, the Willamette Valley, Sussex, Cappadocia, Shiraz, Meknes and Nashik. Cool, rainy, high or tropical places are rare among the training vineyards, and a single 9km cell can miss vineyards a few kilometres away. All six control places, from the Sahara to the Iowa corn belt, fail.

What makes vineyard land

Loss of accuracy when one factor is scrambled, thousandths of AUC

Climate and water Soil



Factors overlap, so a factor's loss understates its total influence
Source: The Signum Average

What makes vineyard land? Scrambling one measure at a time and watching accuracy fall, the most important are night temperature in April, when vines bud and frost kills shoots; winter rain; sunshine in the growing season; irrigation in dry places; and night temperature in the ripening month, which wine-makers associate with fresh acidity and aroma. Soil matters less than climate at this scale, though stony and well-drained soils score better. Because the measures overlap, each loss understates a factor's full influence.

The model has a known bias, and it matters for what follows. It learns where grapes of any kind are grown, not where fine wine is made. In America it rates the Midwest and Texas highly, because hybrid and juice grapes grow there. The quality section below is an attempt to separate the two.

The idle acres

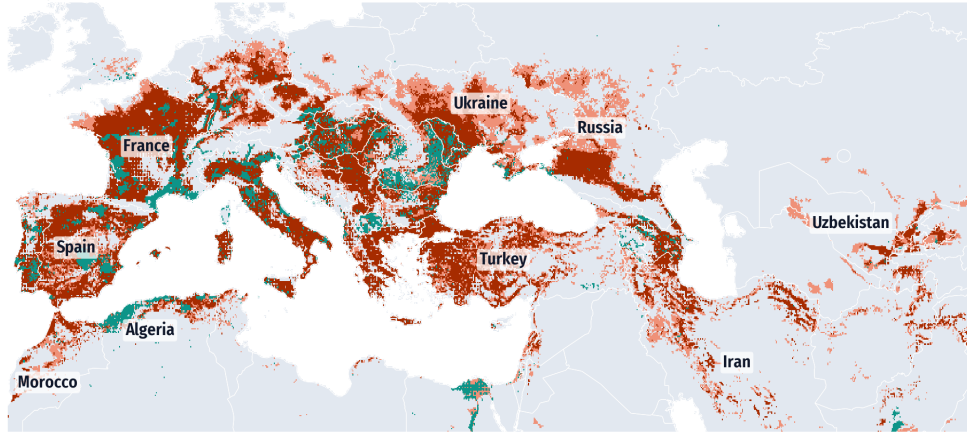
We define two tests. The strict test is passed by land scoring at least as high as the vineyards that hold three-quarters of today's vine hectares in the training countries; the generous test by land as good as those holding nine-tenths. The strict test is the worst case for a newcomer and the generous one the best. Suitable farmland is land that passes, already grows crops and has no vines.

Vineyard land without vineyards

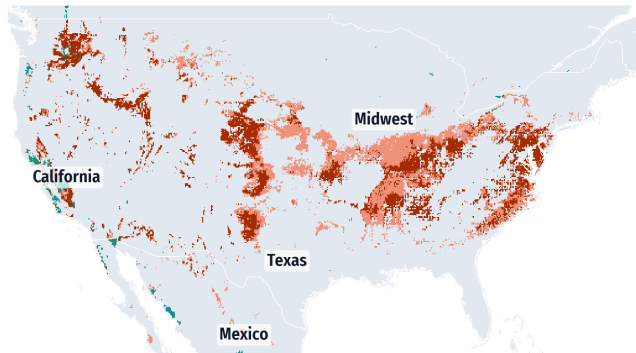
Farmland whose climate and soil match the world's wine regions, by a model that never saw the country it scores. Each cell is about 9 km across; areas are to scale

■ Grapes grown today ■ Suitable, strict test ■ Suitable, generous test

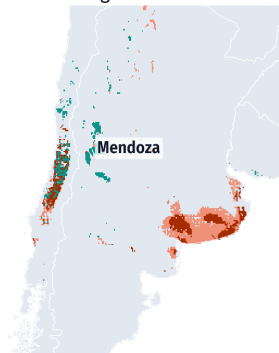
Europe and the Middle East



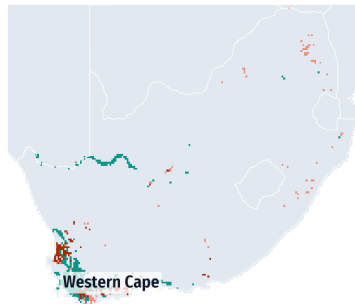
North America



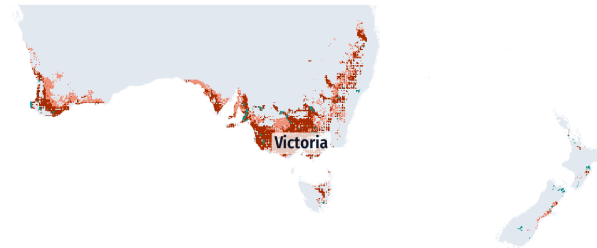
Chile and Argentina



Southern Africa



Australia and New Zealand



Sources: OpenStreetMap contributors; CROPGRIDS (2020); TerraClimate (1991-2020); SoilGrids 2.0; FAO; Natural Earth; The Signum Average

Vineyard land is not scarce. Farmland without vines that passes the strict test covers 115m hectares, and 207m pass the generous test. The world grows wine grapes on 4.5m. Most of the rest grows other crops, as it should: the middle half of established wine countries plant 3-15% of their strict-test farmland with vines.

Where the suitable land is Farmland without vines passing each test, and grapes grown today; thousand hectares

	Strict test	Generous test	All grapes, 2020	Wine grapes, 2023
United States	17,090	39,993	374	236
France	13,524	14,607	766	858
Spain*	9,806	12,296	880	928
Turkey*	8,449	12,133	397	14
Italy*	7,089	7,234	674	650
Ukraine*	6,292	14,637	35	15
Russia	6,144	14,449	75	56
Australia	5,883	10,170	142	145
Iran*	3,863	7,725	174	
Germany	3,541	5,912	101	102
Hungary	3,487	3,933	58	56
Romania	3,447	6,382	160	178
Bulgaria	2,623	3,302	37	54
Greece*	2,360	2,474	97	52
Serbia*	2,171	2,225	20	6

*Not used to train the model. Sources: OpenStreetMap; CROPGRIDS; Anderson, Nelgen and Puga; The Signum Average

Twenty-one countries, each with at least 100,000 hectares of strict-test farmland, plant less than 2.6%, the lower quartile of established wine countries. Together they hold 32% of the world's strict-test farmland and 3% of its wine grapes. Turkey has the most, 8.4m hectares. It grows grapes on about 400,000 hectares, mostly for the table and for raisins, but wine grapes on only 14,000. Ukraine (6.3m hectares) and southern Russia (6.1m), around Krasnodar, Stavropol and Rostov, follow. Iran, where alcohol has been banned for Muslims since 1979, grows grapes on 174,000 hectares and has 3.9m hectares of suitable farmland, much of it in the north-west and along the Zagros. Algeria and Morocco have let their vineyards shrink: between 2000 and 2023 Algeria's wine grapes fell from 30,200 hectares to 2,300 and Morocco's from 49,600 to 6,700. Serbia, Uzbekistan, Azerbaijan and Central Asia complete the list.

The largest unplanted areas Provinces with under 1% of their strict-test farmland under vines; thousand hectares

	Country	Strict test	Generous test	Vines today
Krasnodar	Russia	2,546	3,078	17.6
Victoria	Australia	2,102	3,032	15.1
Buenos Aires	Argentina	1,599	4,855	2.4
Stavropol'	Russia	1,513	1,999	7.5
Colorado	United States	1,410	1,726	0.3
Texas	United States	1,294	2,488	0.7
Indiana	United States	1,069	2,374	0.1
Nebraska	United States	1,023	3,680	0.0
Idaho	United States	962	1,143	0.2
Ohio	United States	884	2,341	0.2
Vinnitsya	Ukraine	845	1,388	2.1
Bayern	Germany	794	1,159	6.8
Kansas	United States	733	3,223	0.0
Tennessee	United States	638	1,046	0.1
Pennsylvania	United States	612	968	5.2

Sources: OpenStreetMap; CROPGRIDS; Natural Earth; The Signum Average

Within established wine countries there is plenty of unplanted land too, from Victoria in Australia to Bavaria. The model's American results deserve scepticism for the reason given above: Texas has a sizeable wine industry, but most of the Midwest's grapes are hybrids.

Who planted, who pulled up Wine-grape area, thousand hectares, in the first recorded year and in 2023

	First year	Area then	Area 2023	Change
United Kingdom	2000	0.9	4.2	+382%
New Zealand	2000	9.9	42.0	+322%
China	2010	29.5	71.2	+141%
Japan	2010	3.7	6.5	+74%
Canada	2000	8.5	13.6	+60%
Czechia	2000	11.3	17.7	+56%
Israel	2000	4.9	6.5	+34%
United States	2000	175.7	235.5	+34%
Spain	2000	1,181.8	928.1	-21%
Serbia	2000	69.0	6.5	-91%
Austria	2000	97.0	43.8	-55%
Romania	2000	222.2	178.2	-20%
Croatia	2000	59.4	16.2	-73%
Morocco	2000	49.6	6.7	-87%

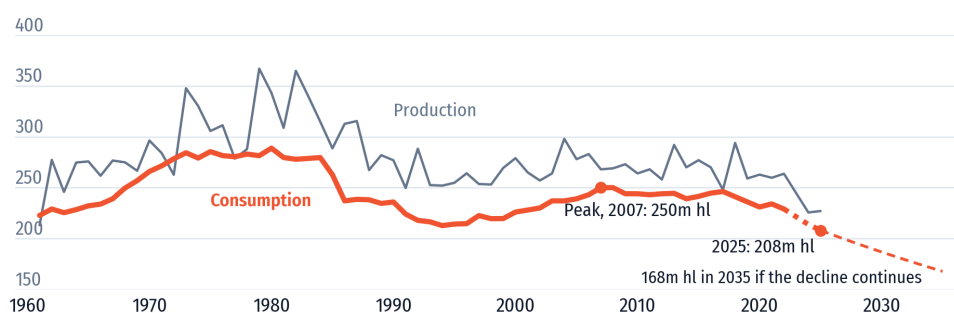
Countries with at least 1,000 hectares in 2023. Countries first recorded after 2000 may have had vineyards earlier. Source: Anderson, Nelgen and Puga (2025)

Where has wine actually spread? England's wine-grape area grew almost fivefold between 2000 and 2023, from 900 hectares to 4,200; New Zealand's quadrupled, to 42,000; China's more than doubled from 2010. Meanwhile Spain pulled up a fifth of its vines, and Serbia, Croatia, Algeria and Morocco most of theirs. The model does not rate Sussex highly. Warming may be outpacing a 30-year climate average, and the training data hold few cool, wet vineyards: both point to the model being too conservative at the cool edge.

A shrinking glass

Drinking less

World wine production and consumption, million hectolitres



2023 not shown: the two sources measure it differently. 2024 derived from the OIV's change for 2025
Sources: Anderson and Pinilla, University of Adelaide (1961-2022); OIV (2024-25); The Signum Average

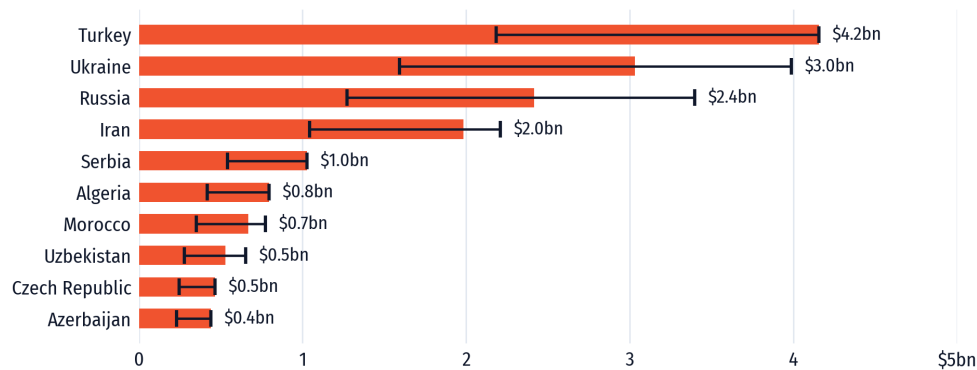
The world drank 208m hectolitres of wine in 2025, according to the OIV, 2.7% less than in 2024 and 14% less than in 2018. In Anderson and Pinilla's series, consumption peaked at 250m hectolitres in 2007 and has never been as low as in 2025 since the series began in 1961. China, France and America account for much of the fall. If the decline of 2018-25, 2.1% a year, continues, the world will drink 168m hectolitres in 2035. Production, at 227m hectolitres in 2025, still exceeds what is drunk; about 30m hectolitres a year go to distilling, vinegar and other uses.

That changes what "forgone" can mean. Planted as densely as the median wine country plants its suitable land, and yielding its median of 57 hectolitres a hectare, the 21 laggards' strict-test farmland would make 121m hectolitres a year, 58% of world consumption; on generous land at the upper-quartile density, three times world consumption. Land is not the constraint; demand is.

So we measure what these countries miss as a share of the market. Suppose each held a share of world sales equal to its share of the world's strict-test farmland. The 21 would then sell about 60m hectolitres more than they do. At the median export price of established producers in 2018-22, US\$2.84 a litre, that is worth \$17bn a year. In the low case, the market shrinks to its 2035 level and the wine sells at the lower-quartile price of \$1.85: \$9bn. In the high case, each country takes the larger of its shares of strict and generous land: \$20bn.

The missing vintages

Wine sales a year that each country forgoes against its share of the world's vineyard-suited farmland, US\$bn at export prices; bars are central, lines low to high



Low: strict-test land, 2035 market if the decline continues, lower-quartile price. High: the larger land share, 2025 market
Sources: OIV; Anderson and Pinilla; Anderson, Nelgen and Puga; OpenStreetMap; The Signum Average

Market share forgone 21 countries that plant wine grapes on under 2.6% of their strict-test farmland

	Share of suitable land	Share of wine grapes	Wine forgone, m hl a year	Sales forgone, \$bn a year (low-high)
Turkey	7.3%	0.30%	14.6	4.16 (2.18-4.16)
Ukraine	5.5%	0.33%	10.7	3.03 (1.59-3.99)
Russia	5.3%	1.24%	8.5	2.42 (1.27-3.40)
Iran	3.4%	0.00%	7.0	1.98 (1.04-2.21)
Serbia	1.9%	0.14%	3.6	1.03 (0.54-1.03)
Algeria	1.4%	0.05%	2.8	0.80 (0.42-0.80)
Morocco	1.3%	0.15%	2.3	0.67 (0.35-0.77)
Uzbekistan	0.9%	0.00%	1.9	0.53 (0.28-0.65)
Czechia	1.2%	0.39%	1.6	0.46 (0.24-0.46)
Azerbaijan	0.7%	0.00%	1.5	0.44 (0.23-0.44)
Slovakia	0.8%	0.26%	1.1	0.31 (0.16-0.31)
Bosnia	0.3%	0.00%	0.7	0.20 (0.10-0.20)
India	0.3%	0.07%	0.6	0.16 (0.08-0.20)
Tajikistan	0.2%	0.00%	0.5	0.15 (0.08-0.15)
Afghanistan	0.2%	0.00%	0.5	0.14 (0.07-0.26)
Pakistan	0.2%	0.00%	0.5	0.13 (0.07-0.43)
Kosovo	0.2%	0.00%	0.4	0.11 (0.06-0.11)
Tunisia	0.3%	0.12%	0.4	0.10 (0.05-0.24)
Kyrgyzstan	0.1%	0.00%	0.3	0.09 (0.05-0.18)
Syria	0.1%	0.00%	0.3	0.07 (0.04-0.09)
Poland	0.1%	0.02%	0.2	0.05 (0.03-0.38)
All 21	31.9%	3.1%	60	17.0 (8.9-20.5)

Sales at export prices. Sources: OIV; Anderson and Pinilla; Anderson, Nelgen and Puga; The Signum Average

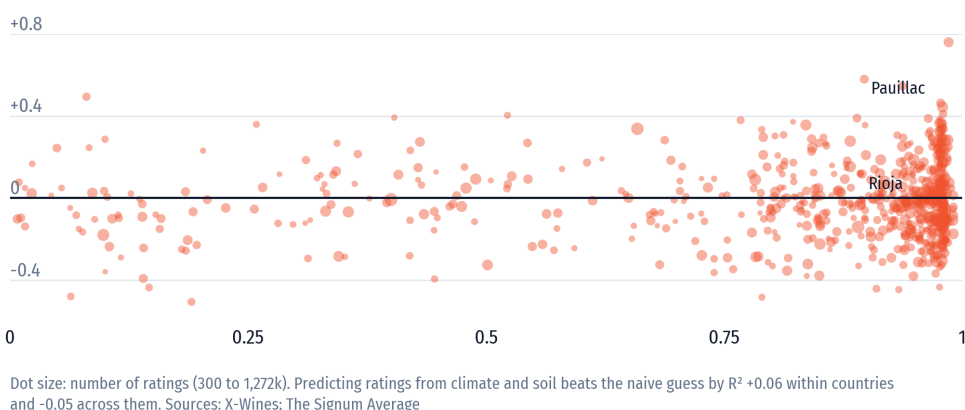
Three caveats temper these numbers. They are valued at export prices, well below what drinkers pay, so they understate the value of a wine industry but overstate what a new one would earn at first. They are not new money for the world: in a market that shrinks every year, every bottle a newcomer sells is one an established producer does not. And they are a benchmark, not a forecast. Land is only one reason wine is made where it is; law, faith, war, sanctions, capital, skills and reputation are others, and in most of these countries they are the binding ones.

Can the land predict quality?

A map of suitable land would be more useful if it could also say how good the wine would be. To test this we used X-Wines: 21m consumer ratings, on a five-star scale, of 100,646 wines rated between 2012 and 2021. We averaged them by region and measured each region against the average for wines of the same type, because sparkling and dessert wines are rated higher than whites; regions with few ratings are pulled towards zero. We located 923 of the 1,204 regions with at least 300 ratings using OpenStreetMap's geocoder and kept the 694 with vines nearby. Each region gets the climate and soil of its vineyards: the average over the grid squares within its outline, weighted by vine hectares.

Good land, not necessarily good wine

694 wine regions: consumer rating against the average for the wine's type, stars (up), and the model's suitability score for their land (across). Rank correlation 0.05



Can the land predict the rating? Regions' ratings against the average for their wine type, predicted from climate and soil

	Better than naive guess (R^2)	Rank correlation	Error, model	Error, naive guess
Out of country	-0.05	0.14	0.186	0.182
Within countries	+0.06	0.30	0.175	0.180

Errors in stars. Naive guess: the average region (out of country) or the country's average (within countries). Source: X-Wines; The Signum Average

The answer is no, or very little. Asked to predict regions in countries it had not seen, a model of climate and soil does no better than guessing the average (R^2 of -0.05 against that guess; rank correlation 0.14). Within countries it does slightly better than the country's own average, explaining about 6% of the variation (rank correlation 0.30). A linear model, or only the 423 regions with at least 2,000 ratings, give the same answer: R^2 between -0.07 and $+0.06$ in every version. Land the suitability model rates highly is not rated more highly by drinkers (rank correlation 0.05).

There are two readings, and both are probably right. The measurement is coarse: squares 9km across cannot see the slope, aspect and gravel that distinguish a great site from its neighbour, and consumer ratings mix producer, price and vintage with place. And quality is made as much as found, by producers, investment and

reputation. Either way we follow our own rule: a model that cannot beat a naive guess is not used to predict. We do not rank the idle land by expected quality. The map says where vines would grow, not how good their wine would be.

What it means

The map shows where vines could go; it does not show where wine would sell. For most of the countries with the most idle vineyard land the obstacles are not agronomic. Iran bans alcohol for Muslims. Ukraine is at war. Russia is under sanctions. Turkey taxes alcohol heavily and restricts its sale. Algeria and Morocco have been pulling up vines for two decades. Where those obstacles ease, the land is there.

For established producers the finding cuts the other way. Their advantage is not land, which is plentiful, but everything else: names, skills, capital and a market that knows them. In a world that drinks less every year, that advantage matters more, not less.

What this analysis cannot tell you

The model finds land where grapes of some kind grow, not where fine wine can be made. It uses average climate for 1991-2020 and so lags behind warming; cool regions such as southern England are probably underrated. Cells are about 9km across, too coarse for slopes, aspect and individual sites, which decide much of a vineyard's quality. The vineyard map is good in Europe, the Americas, South Africa and Oceania and thin in Asia, where the model could not be tested. Sales forgone are a benchmark against land, valued at export prices, and ignore the law, religion, capital, skills and reputation that explain much of where wine is made.

Appendix: data and methods

The grid

All land between 60°S and 72°N on a 5 arc-minute grid (about 9 km at the equator; 2.1m cells). Climate: TerraClimate monthly normals for 1991-2020, averaged from 1/24 degree. From them we compute the growing-season average temperature (April-October, October-April in the south), Winkler degree-days, the Huglin index with its day-length coefficient, night temperature in the ripening month, the coldest month's mean and night temperature, the hottest month's day temperature, the summer-winter and day-night swings, April night temperature (frost at budburst), rain over the year, the growing season, the two months before harvest and the winter, sunshine reaching the ground, the aridity index (rain over potential evaporation) and the growing-season water deficit. Soil at 15-30cm (SoilGrids 2.0): pH, clay, sand, organic carbon, nitrogen, stones and cation-exchange capacity. Irrigation: FAO's Global Map of Irrigation Areas (v5, about 2005), used only where the aridity index is below 0.65; elsewhere irrigation marks intensive farming of any crop and made maize land look like vineyard land. Cropland: the physical area of all 173 crops in CROPGRIDS (2020), capped at the cell's area.

Where vineyards are

Vineyards come from OpenStreetMap (landuse=vineyard), read from Overture Maps' release of September 23rd 2026: 872,520 mapped vineyards covering 3.46m hectares, their areas computed on the ellipsoid and summed by cell. For South Africa this gives 119,887 hectares against 119,471 from a direct OpenStreetMap download. A country's vineyards are taken from OpenStreetMap if it maps at least half of the country's official grape area; otherwise from CROPGRIDS if less than a fifth of its grapes there come from the coarse year-2000 census map, which misplaces vineyards (it shows none at Stellenbosch or Marlborough); otherwise the country is not used for training. A cell counts as vineyard if at least 0.5% of its land is under vines.

The model

Trained on the 26 countries where at least half of the grape area is for wine (Anderson, Nelgen and Puga, 2023 wine-grape area against CROPGRIDS grapes), with at least 1,000 hectares of wine grapes and a trustworthy vineyard map. Countries that grow grapes mainly for the table or raisins, or hardly at all, are never used as evidence against a place; they are only scored. Gradient-boosted trees (scikit-learn HistGradientBoostingClassifier: 300 trees, learning rate 0.06, 31 leaves, L2 penalty 1, balanced classes) on all 6,791 vineyard cells and 400,000 other cells drawn at random. Each training country is scored by models fitted on the other countries (five folds grouped by country); every other country is scored by the average of the five models.

Tests. Hiding one continent at a time, the model separates that continent's vineyard cells from its other land with an area under the ROC curve (AUC) of 0.97 in Europe, 0.93 in North America, 0.90 in South America, 0.95 in Oceania and 0.92 in Africa, against 0.96, 0.87, 0.75, 0.93 and 0.57 for a textbook rule (distance of growing-season temperature from 17°C, the middle of the 12–22°C band in Jones, 2006). Across all wine countries, scored out of country, the AUC is 0.96 against 0.92 for the rule. Asia cannot be tested this way: only 69 vineyard cells, all in Armenia, had a trustworthy map. Of 23 well-known wine or grape regions looked up by name (one 9 km cell each), 13 pass the strict test, three pass only the generous one (Mendoza, Ningxia, the Bekaa) and seven pass neither (Central Otago, Willamette, Sussex, Cappadocia, Shiraz, Meknes and Nashik): cool, rainy, high or tropical places that are rare in the training data, or single cells that miss vineyards a few kilometres away. All six control places (the Sahara, the Amazon, Siberia, Kansas, Iowa, Guangdong) fail both tests.

What matters. Scrambling one factor at a time, the largest losses of accuracy come from April night temperature, winter rain, growing-season sunshine, irrigation in dry land and night temperature in the ripening month. Soil matters less at this resolution; factors overlap, so each loss understates a factor's total influence.

Strict and generous tests

The strict test is passed by land scoring at least as high as the vineyards that hold three-quarters of today's vine hectares in the training countries (score 0.79); the generous test by land scoring as high as those holding nine-tenths (0.49). They cover 4.8% and 8.5% of the training countries' land. Suitable farmland is a cell that passes, has at least 5% cropland and holds no vines; its area is the cell's cropland. Worldwide this is 115m hectares on the strict test and 207m on the generous one, against 4.5m hectares of wine grapes in 2023. The model learns where grapes of any kind grow, not where fine wine is made: it rates the American Midwest highly because hybrid and juice grapes grow there.

Quality

Ratings: X-Wines, 21m consumer ratings (one to five stars, 2012–2021) of 100,646 wines. A region's score is the sum over its ratings of (rating minus the average for that wine type), divided by the number of ratings plus 300, so that sparse regions are pulled to zero. Regions with at least 300 ratings were located with OpenStreetMap's Nominatim geocoder within their country (923 of 1,204); the 694 with mapped vines within their outline are used. A region's climate and soil are averages over grid squares within its outline (at least 0.25 and at most 1 degree from its centre), weighted by vine hectares. Gradient-boosted regression trees (200 trees, 8 leaves, at least 20 regions a leaf), weighted by the log of ratings, were tested in eight folds grouped by country against the average region, and in eight random folds against the country's average. R^2 against the naive guess: -0.05 out of country (rank correlation 0.14) and +0.06 within countries (0.30). Ridge regression and the 423 regions with at least 2,000 ratings give R^2 between -0.07 and +0.06. Because no version beats the naive guess by a useful margin, the quality model is not used to rank land.

Market share and sales forgone

Benchmarks are 20 established wine countries in 2018–22 (Anderson and Pinilla): wine grapes cover 2.6%, 6.1% and 15.4% of their strict-test farmland (lower quartile, median, upper quartile); yields are 35, 57 and 79 hectolitres a hectare; export prices US\$1.85, 2.84 and 3.64 a litre. The 21 countries that plant less than the lower quartile, with at least 100,000 hectares of strict-test farmland, hold 32% of the world's strict-test farmland but 3% of its wine grapes. Planted like wine countries, their land could make 121m hectolitres a year at the median, more than half of what the world drinks, so capacity is not the constraint. We therefore value what each would sell if its share of the world market equalled its share of suitable farmland, minus what it has now. Low case: strict-test land, the market in 2035 if the 2018–25 decline continues (2.1% a year, to 168m hectolitres) and the lower-quartile price. Central: strict-test land, the 2025 market (208m hectolitres) and the median price. High: the larger of the strict and generous land shares, the 2025 market and the median price. Totals: 48, 60 and 72m hectolitres a year, worth US\$8.9bn, 17.0bn and 20.5bn at export prices. This is a benchmark for land left unused, not a forecast: law, religion, capital, skills and reputation decide where wine is made, and new producers would have to take sales from established ones.

Main sources

OpenStreetMap contributors via Overture Maps (ODbL); CROPGRIDS v1.08 (Tang et al., 2024); TerraClimate (Abatzoglou et al., 2018); SoilGrids 2.0 (Poggio et al., 2021); FAO Global Map of Irrigation Areas v5 (Siebert et al., 2013); Natural Earth; K. Anderson, S. Nelgen and G. Puga, Database of Regional, National and Global Winegrape Bearing Areas by Variety, 2000 to 2023 (University of Adelaide, 2025); K. Anderson and V. Pinilla, Annual Database of Global Wine Markets, 1835 to 2024 (University of Adelaide, 2025); OIV, State of the World Wine Sector in 2025 (2026); X-Wines (Azambuja, Morais and Filipe, 2023).

Training countries and vineyard sources

OpenStreetMap vineyards (23): Argentina, Australia, Austria, Belgium, Bulgaria, Canada, Chile, Croatia, Czechia, France, Germany, Hungary, Luxembourg, New Zealand, Portugal, Romania, Russia, Slovakia, Slovenia, South Africa, Switzerland, the United Kingdom and the United States. CROPGRIDS vineyards (3): Albania, Armenia and North Macedonia. Wine countries scored but not trained on, for lack of a trustworthy vineyard map (11): Brazil, Cyprus, Georgia, Greece, Israel, Italy, Lebanon, Moldova, Spain, Turkmenistan and Uruguay. A wine country is one where wine grapes (Anderson, Nelgen and Puga, 2023) are at least half of all grapes (CROPGRIDS, 2020) and cover at least 1,000 hectares.

Choices that were tested and changed

Three versions of the model preceded this one, and each change was made because a test failed. The first used CROPGRIDS for every country's vineyards and WorldClim's 1970-2000 climate; its continent hold-out AUC was 0.56 in Africa and 0.72 in Oceania, because the vineyard map was wrong there. The second used OpenStreetMap and TerraClimate's 1991-2020 climate and included irrigation everywhere; its AUCs rose to 0.90-0.97, but irrigation became its second most important factor and it passed the irrigated maize land of Nebraska and Kansas. The third left irrigation out; Kansas failed, but so did irrigated Mendoza, and the AUCs fell to 0.80-0.96. The published version counts irrigation only in drylands, which keeps the accuracy of the second and the judgement of the third.

Thresholds and areas

Scores are rankings, not probabilities: the model is trained with balanced classes on a sample of non-vineyard land. Thresholds are therefore set by where today's vineyards fall in the ranking, weighted by vine hectares. Suitable farmland is counted as the cropland in a passing cell, not the cell's whole area, so forests, towns and pasture do not count. Countries' totals sum cells by Natural Earth borders; cells on a border go to one country.

Reproducing the results

The code that downloads every input, builds the grid, trains and tests the model and draws the charts is kept in the study's private repository; the data, the suitability map (GeoTIFF) and these methods are public. The scripts fix every random seed, so a rerun gives the same numbers. Vineyards are read from Overture Maps release 2026-09-23.1; a later release will differ slightly as volunteers add vineyards.